SDG BLOCKCHAIN ACCELERATOR

Debugging & Testing Report

SolarVillage: A Hydra-based Energy Settlement & Carbon Issuance Architecture

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SolarVillage – Hydra-Based Energy Settlement & Carbon Issuance Platform

Challenge Owner: UNDP Nigeria
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1. Introduction

This document presents the debugging and validation work conducted on the SolarVillage prototype, a blockchain-based platform designed for microgrid settlement, energy accounting, and carbon credit issuance.

SolarVillage introduces a dual-layer blockchain architecture:

Layer	Purpose
Hydra Layer	High-frequency energy settlement and operational accounting
Cardano Layer 1	Carbon asset issuance and long-term audit registry

Testing activities were performed to ensure that the system behaves deterministically, securely, and auditable across both layers.

Unlike traditional applications where testing primarily focuses on API responses or UI functionality, blockchain systems require validation of:

- deterministic state transitions
- validator correctness
- replay and double-spend protection
- cryptographic integrity of data commitments
- interoperability between on-chain and off-chain components

The debugging process therefore focused on validating the full lifecycle of the SolarVillage system, from meter data ingestion through carbon credit issuance.

2. Testing Objectives

The testing program aimed to validate five critical properties of the system.

2.1 Deterministic Energy Settlement

Energy settlement must produce the same state transition when replayed from the same inputs. This ensures that auditors and verifiers can reconstruct system state independently.

2.2 Accurate Energy Accounting

All watt-hours delivered must be correctly accounted for through the minting and retirement of **WH audit tokens**, which act as the bridge between energy delivery and carbon issuance.

2.3 Secure Carbon Credit Issuance

Carbon credits (COG) must only be issued when:

- verified energy has been delivered
- corresponding WH tokens have been retired
- a valid Period Receipt has been consumed.

2.4 Double-Counting Prevention

The system must ensure that:

- energy units cannot be reused across multiple issuance periods
- receipts cannot be consumed more than once
- settlement transactions cannot be replayed.

2.5 Operational Stability

The Hydra settlement layer must support high-frequency transaction flows typical of microgrid environments.

3. Testing Environment

Testing was conducted in an environment designed to approximate production deployment conditions.

3.1 Blockchain Infrastructure

Component	Environment
Cardano Network	Preprod Testnet
Hydra Node	v1.1.x
Cardano Node	Conway-compatible release
Smart Contracts	Aiken

Hydra heads were configured to simulate a single microgrid settlement environment.

3.2 Off-Chain Infrastructure

Several off-chain services were used to simulate the operational environment of a microgrid.

Service	Role
Metering Gateway	Produces signed meter readings
Tariff Oracle	Publishes tariff updates
Backend Service	Constructs settlement transactions
Dashboard	Displays operational metrics

These services coordinate the system but **do not control on-chain outcomes**.

4. Smart Contract Validation

SolarVillage smart contracts were implemented in **Aiken** and deployed across both Hydra and Cardano L1.

The following validator scripts were tested:

Validator	Role
Consumer Vault	Holds custodial consumer balances

Settlement Validator	Applies tariff and metering logic
Grid WH Vault	Stores accumulated energy tokens
Period Receipt Validator	Commits issuance data
COG Minting Policy	Issues carbon credits
Certificate NFT Policy	Creates issuance certificates

Each validator enforces explicit state transitions using redeemers and datum structures.

5. Settlement Testing

Settlement testing focused on validating the 15-minute operational accounting cycle.

Each settlement interval includes:

1. ingestion of meter readings
2. tariff lookup
3. consumer balance update
4. operator revenue allocation
5. WH token minting

All of these operations occur in a **single Hydra transaction**.

Settlement Transaction

Below is a simplified example used during testing.

Consumer Vault State Before Settlement

```
{
 "consumer_id": "C001",
 "grid_id": "GRID-NG-01",
 "svn_balance": 5000
}
```

Meter Reading Input

```
{
 "meter_id": "MTR-001",
```

```
 "wh_consumed": 750,
 "interval": "15min"
}
```

Tariff

```
{
 "tariff_rate": 0.2,
 "unit": "SVN/kWh"
}
```

Transaction Construction

```
cardano-cli transaction build \
 --tx-in <consumer-vault-utxo> \
 --tx-out <operator-wallet> \
 --tx-out-inline-datum-file updated-consumer-datum.json \
 --out-file settlement.raw
```

Expected Outcome

State Change	Result
Consumer Balance	5000 → 4850 SVN
Operator Revenue	+150 SVN
Grid WH Vault	+750 WH

Output

```
{
 "status": "success",
 "consumer_balance_after": 4850,
 "operator_revenue": 150,
 "wh_minted": 750
}
```

6. Carbon Issuance Testing

Carbon issuance occurs periodically and is anchored on Cardano Layer 1.

The issuance process includes:

1. WH retirement
2. creation of a Period Receipt
3. receipt decommit to L1
4. minting of carbon tokens

Issuance Calculation example

WH retired = 182,500
Emission factor = 0.42 gCO_2e / Wh

Result:

- COG minted = 76,650 grams CO_2e

Issuance Transaction

```
cardano-cli transaction build \
--tx-in <period-receipt-utxo> \
--mint "76650 COG" \
--out-file carbon-issuance.raw
```

Expected Output

```
{
 "status": "success",
 "cog_minted": 76650,
 "certificate_nft": true
}
```

7. Negative-Path Testing

Several adversarial scenarios were tested to confirm system resilience.

Invalid Redeemer

Attempting to submit a settlement transaction with an incorrect redeemer.

Example:

```
{
"redeemer": "Create"
}
```

Result:

- Transaction rejected
- Reason: unexpected redeemer

Duplicate Period Receipt

Attempting to create a second receipt for the same `(grid_id, period_id)`.

Result:

- Rejected by PeriodReceiptValidator

Receipt Replay Attempt

Attempting to reuse a consumed receipt to mint carbon credits again.

Result:

Rejected due to consumed UTxO

8. Hydra Runtime Validation

Hydra runtime logs were monitored during testing to confirm head lifecycle behavior.

log:

```
{
 "tag": "HeadIsOpen",
 "headId": "14cab222dcef7584fba3b630a1eaef996786fa80555511f836faf4cc"
}
```

```
Settlement event log:
{
 "event": "SettlementApplied",
 "consumer_id": "C001",
 "wh_consumed": 750
}
```

These logs confirm that settlement events were executed within Hydra before anchoring on Layer 1.

9. Issues Discovered During Debugging

Issue	Description	Root Cause	Resolution	Status
Settlement balance mismatch	Early settlement tests produced incorrect operator revenue totals	Tariff rounding logic differed between off-chain and validator calculations	Standardized tariff rounding to validator logic	Resolved
Receipt duplication	Prototype initially allowed creation of duplicate receipts	Missing `(grid_id, period_id)` uniqueness check	Added validator check enforcing unique receipt	Resolved
Carbon issuance mismatch	Initial carbon issuance produced incorrect COG totals	Methodology factor applied incorrectly	Corrected emission factor conversion logic	Resolved
Hydra decommit delay	Some receipt decommit transactions experienced delays	Hydra head synchronization timing	Introduced decommit retry logic	Mitigated
Meter batch validation	Malformed meter batch signatures initially accepted	Missing signature verification step	Added gateway signature verification	Resolved

10. Performance Observations

Testing showed that the Hydra-based operational layer performs efficiently under simulated workloads.

Key observations:

- settlement transactions complete within milliseconds inside Hydra
- L1 interactions remain limited to periodic anchoring
- operational accounting scales independently of Cardano L1 throughput.

This architecture significantly reduces transaction costs and improves scalability for microgrid environments.

11. Limitations of Current Testing

As this work represents a prototype deployment, several limitations remain.

Limited Multi-Grid Testing

Testing simulated only a small number of microgrid environments.

Metering Hardware Assumptions

The system assumes trusted metering hardware. Hardware tampering was not simulated.

Hydra Stress Testing

Large-scale Hydra head stress testing was outside the scope of this phase.

12. Next Testing Steps

Future testing will focus on:

- Hydra stress testing with large numbers of meters
- independent smart contract audit
- registry interoperability testing
- validator execution cost optimization.